

INDIAN STATISTICAL INSTITUTE

Mid-Semester Examination

Second semester 2025-2026

M. Math (Second year)

Symplectic Geometry

Date: 24 February, 2026

Maximum Marks: 40

Duration: 2 hours

Answer all questions.

If you use any result then state it clearly.

- (1) Let $\mathbb{T}^n$ denote the n -dimensional torus which is an n -fold product of circles.
- (a) Show that $\mathbb{T}^{2n}$ carries a symplectic form.
 - (b) Find two symplectic forms on $\mathbb{T}^{2n}$ which are not symplectomorphic.
 - (c) Find two symplectic forms on $\mathbb{T}^2$ which are not isotopic but symplectomorphic.
 - (d) Find two symplectic forms on $\mathbb{T}^2$ which are isotopic. 14

- (2) Let (M, ω) be a symplectic manifold and $f \in C^\infty(M, \mathbb{R})$. Prove the following:
- (a) The integral curves of the Hamiltonian vector field X_f lies in the level sets of f .
 - (b) Prove that $X_{f \circ \varphi^{-1}} = \phi_* X_f$ for all symplectomorphism φ of (M, ω) . 12

[If X is a vector field on M and φ is a diffeomorphism of M , then $\varphi_* X$ is a vector field defined by $(\varphi_* X)\varphi(p) = d\varphi_p(X(p))$ for all $p \in M$].

- (3) Let M be a connected, closed ^{or}orientable n -manifold and ω_0, ω_1 be two volume forms on M (compatible with the orientation on M). Suppose that the ~~symplectic~~ volume of (M, ω_0) is the same as that of (M, ω_1) ; that is,

$$\int_M \omega_0 = \int_M \omega_1.$$

- (a) Define $\omega_t = (1-t)\omega_0 + t\omega_1$, $0 \leq t \leq 1$. Show that ω_t is a volume form for all $t \in [0, 1]$.
- (b) Prove that there exists an isotopy δ_t , $0 \leq t \leq 1$, such that $\delta_t^* \omega_t = \omega_0$ for all $t \in [0, 1]$. 10

(Assume de-Rham's theorem: If M is a closed orientable n -manifold then $\phi : H_{deR}^n(M) \rightarrow \mathbb{R}$ defined by $\phi([\sigma]) = \int_M \sigma$ is an isomorphism, where σ is any n -form on M).

- (4) Let $\mathbb{S}^n$ denote the n -sphere in $\mathbb{R}^{n+1}$.
- (a) Find a symplectic form on $\mathbb{S}^2$. What are the closed Lagrangian submanifolds of $\mathbb{S}^2$?
 - (b) Show that $\mathbb{S}^{2n}$ does not admit a symplectic form for $n > 1$. Give a detailed answer. 12